

```

• ▶ python3 server.py
🚀 TLS Private Routing & FTP Server Hub Is Active...
👤 [JOINED] dina connected from ('127.0.0.1', 39260)
👤 [JOINED] minha connected from ('127.0.0.1', 39274)
|

```

Encrypted data:

```

0x0010: 003c 7d06 4000 4006 bfb3 7f00 0001 7f00 .<}.@.@.....
0x0020: 0001 dfa0 ff98 9f9a 1ccf 0000 0000 a002 .....
0x0030: ffd7 fe30 0000 0204 ffd7 0402 080a 8d5d ...0.....]
0x0040: 4806 0000 0000 0103 030a H.....
18:34:05.105588 IP 127.0.0.1.65432 > 127.0.0.1.57248: Flags [S.], seq 3170372476, ack 2677677264, w
in 65483, options [mss 65495,sackOK,TS val 3615512429 ecr 2371700742,nop,wscale 10], length 0
0x0000: 0000 0000 0000 0000 0000 0000 0800 4500 .....E.
0x0010: 003c 0000 4000 4006 3cba 7f00 0001 7f00 .<..@.@.<.....
0x0020: 0001 ff98 dfa0 bcf8 0b7c 9f9a 1cd0 a012 .....|.....
0x0030: ffc0 fe30 0000 0204 ffd7 0402 080a d780 ...0.....
0x0040: 576d 8d5d 4806 0103 030a Wm.]H....
18:34:05.105605 IP 127.0.0.1.57248 > 127.0.0.1.65432: Flags [.] , ack 1, win 64, options [nop,nop,TS
val 2371700742 ecr 3615512429], length 0
0x0000: 0000 0000 0000 0000 0000 0000 0800 4500 .....E.
0x0010: 0034 7d07 4000 4006 bfb3 7f00 0001 7f00 .4}.@.@.....
0x0020: 0001 dfa0 ff98 9f9a 1cd0 bcf8 0b7d 8010 .....}..
0x0030: 0040 fe28 0000 0101 080a 8d5d 4806 d780 .@.(.....]H...
0x0040: 576d Wm
18:34:05.106171 IP 127.0.0.1.57248 > 127.0.0.1.65432: Flags [P.], seq 1:1526, ack 1, win 64, option
s [nop,nop,TS val 2371700742 ecr 3615512429], length 1525
0x0000: 0000 0000 0000 0000 0000 0000 0800 4500 .....E.
0x0010: 0629 7d08 4000 4006 b9c4 7f00 0001 7f00 .)}.@.@.....
0x0020: 0001 dfa0 ff98 9f9a 1cd0 bcf8 0b7d 8018 .....}..
0x0030: 0040 041e 0000 0101 080a 8d5d 4806 d780 .@.....]H...
0x0040: 576d 1603 0105 f001 0005 ec03 0313 684b Wm.....hK

```

Client1:

```
• ▶ python3 client.py
👤 Apna Username likhein: minha
✅ Welcome minha! Private Chat: @user msg | File: @user /send filename

🔒 [Private from dina]: hello
🗨️ You: @dina hello
🗨️ You: |
```

Client2:

```
• ▶ python3 client.py
👤 Apna Username likhein: dina
✅ Welcome dina! Private Chat: @user msg | File: @user /send filename

🗨️ You: @minha hello
🔒 [Private from minha]: hello
🗨️ You: |
```

Network Capture:

Apply a display filter ... <Ctrl-/>							
Time	Delta	Source	Destination	Protocol	Length	Info	
26 8.260286984	0.000025981	127.0.0.1	127.0.0.1	TCP	74	65432 → 54092 [SYN, ACK] Seq=0 Ack=1 Win=65483 Len=0 MSS=65495 SACK_PERM TSval=1550139078 TSecr=3923011438 WS=102	
27 8.260307594	0.000020610	127.0.0.1	127.0.0.1	TCP	66	54092 → 65432 [ACK] Seq=1 Ack=1 Win=65536 Len=0 TSval=3923011438 TSecr=1550139078	
28 8.260987079	0.000679485	127.0.0.1	127.0.0.1	TLSv1.3	1591	Client Hello (SNI=localhost)	
29 8.261000634	0.000013555	127.0.0.1	127.0.0.1	TCP	66	65432 → 54092 [ACK] Seq=1 Ack=1526 Win=79872 Len=0 TSval=1550139079 TSecr=3923011439	
30 8.278386335	0.017385701	127.0.0.1	127.0.0.1	TLSv1.3	3241	Server Hello, Change Cipher Spec, Application Data, Application Data, Application Data, Application Data	
31 8.278402172	0.000015837	127.0.0.1	127.0.0.1	TCP	66	54092 → 65432 [ACK] Seq=1526 Ack=3176 Win=97280 Len=0 TSval=3923011457 TSecr=1550139097	
32 8.279458910	0.001056738	127.0.0.1	127.0.0.1	TLSv1.3	146	Change Cipher Spec, Application Data	
33 8.279706574	0.000247664	127.0.0.1	127.0.0.1	TLSv1.3	321	Application Data	
34 8.321399359	0.041692785	127.0.0.1	127.0.0.1	TCP	66	54092 → 65432 [ACK] Seq=1606 Ack=3431 Win=100352 Len=0 TSval=3923011500 TSecr=1550139098	
35 8.321424600	0.000025241	127.0.0.1	127.0.0.1	TLSv1.3	362	Application Data, Application Data	
36 8.321433129	0.000008529	127.0.0.1	127.0.0.1	TCP	66	54092 → 65432 [ACK] Seq=1606 Ack=3727 Win=100352 Len=0 TSval=3923011500 TSecr=1550139140	
37 11.994230507	3.672797378	127.0.0.1	127.0.0.1	TLSv1.3	92	Application Data	
38 11.994677926	0.000447419	127.0.0.1	127.0.0.1	TLSv1.3	159	Application Data	
39 11.994699541	0.000021615	127.0.0.1	127.0.0.1	TCP	66	59206 → 65432 [ACK] Seq=1632 Ack=3820 Win=100352 Len=0 TSval=2009247092 TSecr=148093620	
40 16.709004686	4.714305145	127.0.0.1	127.0.0.1	TLSv1.3	92	Application Data	
41 16.709464164	0.000459478	127.0.0.1	127.0.0.1	TLSv1.3	159	Application Data	
42 16.709492225	0.000028061	127.0.0.1	127.0.0.1	TCP	66	54076 → 65432 [ACK] Seq=1632 Ack=3820 Win=100352 Len=0 TSval=2113034990 TSecr=918358152	
43 10.455440767	9.044157522	127.0.0.1	127.0.0.1	TLSv1.3	92	Application Data	

Frame 28: Packet, 1591 bytes on wire (12728 bits), 1591 bytes captured (12728 bits) on interface lo, id Ethernet II, Src: 00:00:00:00:00:00 (00:00:00:00:00:00), Dst: 00:00:00:00:00:00 (00:00:00:00:00:00) Internet Protocol Version 4, Src: 127.0.0.1, Dst: 127.0.0.1 Transmission Control Protocol, Src Port: 54092, Dst Port: 65432, Seq: 1, Ack: 1, Len: 1525 Transport Layer Security

0000 45 00E-
0010 06 29 be 04 40 00 40 06 78 c8 7f 00 00 01 7f 00@-@ x.....
0020 00 01 d3 4c ff 98 40 4f 65 0c b5 d3 23 dc 80 18L-@ e.....
0030 00 40 04 1e 00 00 01 01 08 0a e9 d4 67 6f 5c 65@-.....go\.....
0040 3e c6 16 03 01 05 f0 01 00 05 ec 03 03 1c 85 30>.....0.....
0050 5c 54 5a df 41 a0 68 86 f9 5f ce d6 88 2e fb 75 \TZ A h.....u
0060 b1 58 1b dc d4 5d 6c 4f b2 ba c2 06 0c 20 f9 91 X-...]]l0.....
0070 cc ab 1c 92 56 d1 21 b5 23 c8 b9 e8 dc 83 2a e2V!:-#.....
0080 04 ab 7e 96 03 eb 80 b9 62 29 23 c5 34 b8 00 22b)#-4-.....
0090 13 02 13 03 13 01 c0 2c c0 30 c0 2b c0 2f cc a9-0-#-/-.....
00a0 cc a8 c0 24 c0 28 c0 23 c0 27 00 9f 00 9e 00 6b\$(#-'k.....
00b0 00 67 01 00 05 81 ff 01 00 01 00 00 00 00 00g.....
00c0 0c 00 00 09 6c 6f 63 61 6c 68 6f 73 74 00 00loca lhost.....
00d0 02 01 00 00 0a 00 12 00 10 11 ec 00 1d 00 17 00
00e0 1e 00 18 00 19 01 00 01 01 00 23 00 00 00 16 00#.....
00f0 00 00 17 00 00 00 0d 00 36 00 34 09 05 09 06 096-4.....
0100 04 04 03 05 03 06 03 08 07 08 08 08 1a 08 1b 08
0110 1c 08 09 08 0a 08 0b 08 04 08 05 08 06 04 01 05
0120 01 06 01 03 03 03 01 03 02 04 02 05 02 06 02 00
0130 2b 00 05 04 03 04 03 03 00 2d 00 02 01 01 00 33+.....3.....